

Homework — 1.1.1 Structure and Function of the Processor

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.1.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Complete the table by filling in the **full name** and **role** of each register. [4] (AO1)

Register	Full name	Role
PC	(a)	(b)
ACC	(c)	(d)
MAR	(e)	(f)
CIR	(g)	(h)

2. The events of the **fetch** and **decode** stages of the fetch–decode–execute cycle are shown below in a jumbled order. Number them **1–5** to show the correct order. [2] (AO1)

Event	Order (1–5)
The instruction is transferred from memory to the MDR along the data bus	
The Control Unit decodes the instruction in the CIR, splitting it into opcode and operand	
The address held in the PC is copied into the MAR	
The contents of the MDR are copied into the CIR	
The PC is incremented by 1	

3. A processor uses the instruction set and memory contents below. The PC starts at **0** and the ACC starts at **0**. Complete the trace table, writing the value held in each column **after each instruction has finished executing**. The trace stops when HALT is fetched. [3] (AO2)

Instruction	Meaning
LOAD n	Copy the contents of memory address n into the ACC
ADD n	Add the contents of memory address n to the ACC
STORE n	Copy the contents of the ACC into memory address n
HALT	Stop execution

Address	Contents
0	LOAD 4
1	ADD 5
2	STORE 6
3	HALT
4	7
5	5
6	0

PC	CIR	ACC	Contents of address 6

4. Complete the passage about processor performance using words from the word bank. Not all words are used. **[3]** (AO1)

Several factors affect the performance of a processor. Increasing the (a) _____ speed increases the number of fetch–decode–execute cycles completed each second. (b) _____ is a small amount of very fast memory built into or close to the CPU; it holds frequently used instructions and data so the processor makes fewer slow accesses to (c) _____. With (d) _____, the processor overlaps the stages of the cycle: while one instruction is being (e) _____, the next is being decoded and a third is being fetched. A processor with more than one (f) _____ can carry out several instructions truly in parallel.

Word bank: cache, clock, core, executed, interrupt, pipelining, RAM, register, ROM

5. A processor in an embedded system has a **16-bit address bus** and an **8-bit data bus**.

(a) Calculate the maximum number of memory locations that can be directly addressed. Show your working. **[1]** (AO2)

(b) State **one** effect on the performance of the processor of widening the data bus to 16 bits. **[1]** (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

6. Match each bus to its description by writing the correct **letter** in the Match column. Two descriptions are not needed. **[3]** (1.1.1)

Bus	Match
Address bus	
Data bus	
Control bus	

Letter	Description
A	Bidirectional; carries data and instructions between the CPU and memory
B	Unidirectional; carries the address of the memory location to be accessed, from the CPU to memory
C	Bidirectional; carries signals such as read/write, clock and interrupt
D	Unidirectional; carries the results of calculations back to the ACC
E	Bidirectional; carries the address of the next instruction from memory to the CPU

7. Explain what **cache** memory is and **one** reason it improves CPU performance. **[2]** (1.1.1)

8. State what is meant by **pipelining** in a processor. **[1]** (1.1.1)